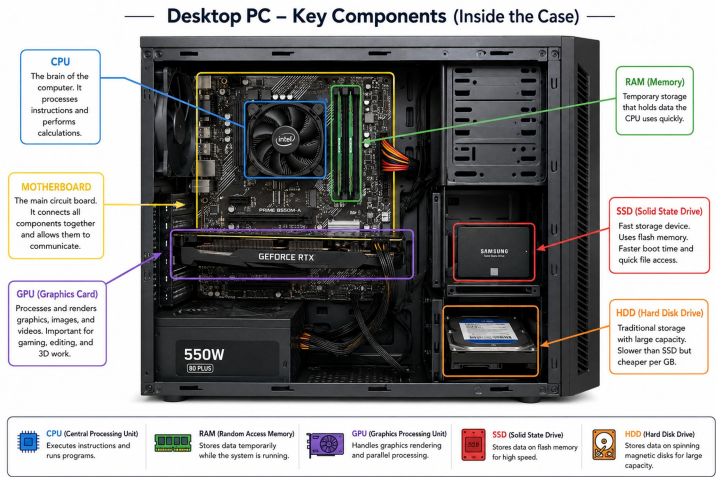


HARDWARE COMPONENT THEORY

1. Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.

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2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).

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Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Full Form	Hard Disk Drive	Solid State Drive
Storage Technology	Uses spinning magnetic disks	Uses flash memory chips
Speed	Slower (Boot: 30–60 seconds)	Much faster (Boot: 10–20 seconds)
Read/Write Speed	Around 80–160 MB/s	Around 500–7000 MB/s (depending on SATA or NVMe SSD)
Durability	Less durable due to moving parts	More durable because it has no moving parts
Shock Resistance	Low – can be damaged by drops or shocks	High – resistant to shocks and vibrations
Noise	Makes noise while operating	Silent operation
Power Consumption	Uses more power	Uses less power
Weight	Heavier	Lighter
Price	Cheaper per GB	More expensive per GB
Typical Use Cases	Desktop PCs, file storage, backups, CCTV systems	Laptops, gaming PCs, office computers, high-performance systems
Best For	Large storage at a lower cost	Fast booting, quick application loading, gaming, and multitasking

3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

=> **1. CPU (Central Processing Unit)**

The CPU is the brain of the computer. It processes game instructions, controls player movements, manages AI, physics, and communicates with all other hardware to keep the game running smoothly.

2. RAM (Random Access Memory)

RAM temporarily stores the game data that the CPU needs while the game is running. More RAM allows the game to load faster, switch between tasks smoothly, and reduces lag caused by insufficient memory.

3. GPU (Graphics Processing Unit)

The GPU is responsible for rendering the game's graphics, including characters, maps, lighting, and visual effects. A powerful GPU provides higher frame rates (FPS), better image quality, and a smoother gaming experience in games like PUBG and Free Fire.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

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Hardware Component	Why Upgrade It?
Graphics Card (GPU)	The GPU has the biggest impact on gaming performance. A better graphics card gives higher FPS, smoother gameplay, and improved graphics quality.
Processor (CPU)	A faster CPU helps the game process player movements, AI, physics, and background tasks without slowing down. It also reduces CPU bottlenecks.
RAM	Upgrade to at least 16 GB RAM for smooth gameplay and multitasking. More RAM reduces lag and improves game loading performance.
SSD (Solid State Drive)	Installing FIFA on an SSD greatly reduces loading times and helps the game start faster than an HDD.
Power Supply (PSU)	If you install a more powerful GPU or CPU, you may need a higher-wattage PSU to provide stable power.
Cooling System	Better air or liquid cooling keeps the CPU and GPU at safe temperatures, preventing overheating and performance drops during long gaming sessions.
Motherboard (if required)	If the new CPU or RAM is not compatible with your current motherboard, you may need to upgrade it to support newer hardware.